



AI-driven control algorithm using machine learning and genetic optimization for enhancing visual comfort in adaptive façades

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ABSTRACT

Effective management of daylight and visual comfort in office spaces remains a challenge, as existing shading systems often lack adaptability to changing environmental conditions and occupant needs. This paper presents an AI-driven real-time shading control algorithm that optimizes visual comfort using machine learning-based surrogate models and evolutionary optimization. A non-conventional adaptive façade was simulated using Radiance and Ladybug Tools across nine U.S. climates. Four machine learning models were evaluated for predicting Task Illuminance (E_t) and Vertical Eye Illuminance (E_v), with Extra Trees achieving the highest accuracy ($R^2 = 0.95$). A Non-dominated Sorting Genetic Algorithm II (NSGA-II) balances glare reduction and daylight utilization by optimizing façade configurations in real time. In contrast to prior approaches constrained to fixed geometries and single-objective control, this paper introduces a generalizable multi-objective control framework. Results show that AI-driven optimization significantly improves adaptive façade performance, offering a scalable solution for intelligent daylight and comfort management.

1. Introduction

The amount of daylight entering through windows plays a crucial role in minimizing glare while ensuring adequate levels of daylight to enhance the health and visual comfort of occupants, as well as to reduce electricity consumption [1]. Various shading systems, such as solar shades, Venetian blinds, roller blinds, and static shadings, have been developed to manage natural light. Static or fixed shading systems, which do not adapt to environmental changes throughout the day or across seasons, are generally less preferred due to their rigidity. However, even dynamic systems are frequently used ineffectively.

Several challenges are associated with manual shading control. First, constant user attention is required, as frequent adjustments are necessary to maintain optimal thermal or visual conditions. Furthermore, manual adjustments often fail to reflect real-time environmental changes, resulting in inappropriate configurations such as leaving shades open during high solar gain or closed on overcast days. Improper

shading control can also increase cooling loads during summer while reducing access to natural outdoor views, which can affect the well-being and comfort of occupants. In shared spaces, control over shading systems is often unequal, with individuals nearest the controls adjusting shading based on personal preferences, sometimes disregarding the comfort of others. Intelligent occupant-centric shading controls have emerged as an innovative solution to manage natural light and improve energy efficiency in buildings. These systems use sensors to monitor environmental factors such as sunlight intensity, temperature, and occupancy [2]. Based on the data collected, the system automatically adjusts shading devices to optimize indoor lighting and thermal conditions [3,4]. However, such sensor-based systems also have limitations in meeting occupant comfort needs. The integration of adaptive shading façades that account for both occupant position and the sun's trajectory presents a valuable opportunity to enhance visual comfort. However, existing systems often fall short in key areas such as real-time

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Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
ANN	Artificial Neural Network
AP	Attractor Point
ASE	Annual Sunlight Exposure
cDA	Continuous Daylight Autonomy
DA	Daylight Autonomy
DGP	Daylight Glare Probability
DGPs	Simplified Daylight Glare Probability
E_v	Vertical Eye Illuminance
E_t	Task Illuminance
GA	Genetic Algorithm
HB	Honeybee
LB	Ladybug
LightGBM	Light Gradient Boosting Machine
MAE	Mean Absolute Error
MBC	Model-Based Control
MSE	Mean Squared Error
MPC	Model-Based Predictive Control
NSGA-II	Non-dominated Sorting Genetic Algorithm II
RF	Random Forest
R^2	Coefficient of Determination (R-Squared)
sDA	Spatial Daylight Autonomy
UDI	Useful Daylight Illuminance

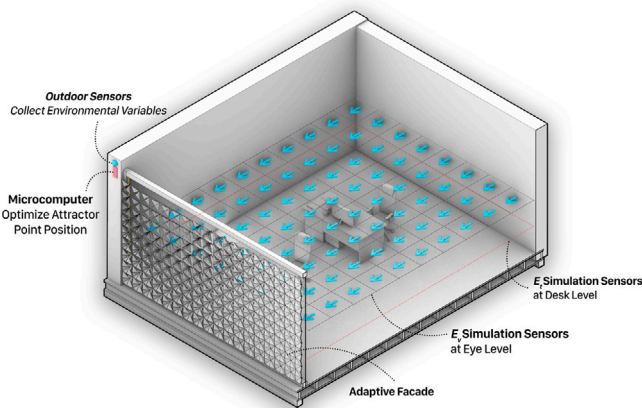


Fig. 1. 3D section of the model configuration.

responsiveness, model generalizability, and adaptability to complex building geometries. Addressing these gaps is essential for developing scalable and intelligent control systems capable of improving occupant experience across diverse architectural contexts.

By integrating machine learning-based predictive control models, this research aims to propose a method for adaptive façade design by developing a control algorithm that dynamically responds to both environmental conditions and occupant location, as depicted in Fig. 1. The proposed method is designed to be generalizable across different climates, building types, and, importantly, can be applied to façades with complex geometries. To enhance performance, a multi-objective genetic optimization algorithm is employed to fine-tune the model for maximum visual comfort. While this approach addresses key limitations of existing shading systems, such as inflexible control logic and limited adaptability, it does not yet incorporate weather forecasts or real-time occupant tracking features identified as directions for future enhancement.

This paper is structured as follows: Section 2 presents a review of related work in real-time shading control systems and algorithms using

machine learning. Section 3 describes the methodology, including data collection, model architecture, and the genetic optimization process. Section 4 presents the results and evaluates the performance of the proposed system across different scenarios. Section 5 discusses the findings in greater detail, and finally, Section 6 concludes the paper by highlighting the research limitations and outlining directions for future work.

2. Literature review

The field of building control systems has garnered increasing attention in recent years, with a growing body of research dedicated to developing control algorithms that enhance both energy efficiency and occupant comfort. Control strategies in this domain are generally categorized into three main types: threshold controllers, blocking controllers, and mode and scene controllers [5]. Threshold-based systems typically activate shading devices based on exterior solar radiation levels; however, such approaches have been found to be insufficient in effectively minimizing energy demand [6]. Blocking controllers improve upon this by adjusting shading positions according to the sun's trajectory, thereby helping to reduce glare and maintain appropriate illuminance levels, while also lowering lighting and cooling loads compared to static systems [7,8]. Mode and scene controllers represent a more advanced class, combining inputs such as vertical irradiance and indoor requirements to manage shading based on specific user modes or activities.

New computer-aided approaches, such as machine learning and genetic algorithms, are increasingly applied in buildings to manage multi-functionality and non-linear adaptations, making them useful in comparative design [9,10]. Automated control strategies (without user intervention) and occupant-centric control strategies (with user interventions) are two methods for controlling adaptive façades [11]. The third new method was developed as a result of the combination of the first two strategies. In this approach, intelligent occupant-centric shading controls rely primarily on artificial intelligence. These forms of predictive controls are a combination of hardware and software that enable an independent operation of the shading system through a learning process [12]. Böke et al. [13] emphasized the need for intelligent control systems capable of making decisions based on current demands and environmental information.

Wang et al. [14] proposed an occupant-centric adaptive façade system that leverages deep learning to detect posture-related indicators of glare and thermal discomfort in real time. By training a convolutional neural network (CNN) on labeled occupant postures extracted via the OpenPose algorithm, the system could recognize discomfort with high accuracy (91.67%) and adjust the façade and HVAC systems accordingly. This non-contact method allows for personalized environmental control without the need for wearable sensors, representing a significant advancement in adaptive comfort systems for shared spaces.

Similarly, Luo et al. [15] introduced a machine learning-based shading controller for open-plan offices using radial basis function neural networks. Their model used real-time daylight simulations and surrogate modeling techniques to optimize vertical eye illuminance across multiple occupant positions. The approach addressed challenges associated with spatially and temporally varying occupancy, demonstrating superior performance in balancing visual comfort, daylight autonomy, and seasonal heat gains compared to traditional fixed or manually operated blinds.

In a subsequent study, Luo et al. [16] further refined this approach by identifying the most influential control variables for ML-assisted louver systems. Through a comprehensive feature selection process applied to simulation-based datasets, they found that spatiotemporal occupancy data played the most critical role in ensuring accurate prediction of visual comfort indicators such as vertical illuminance. Their work emphasized the importance of variable selection in building efficient and responsive predictive models for automated shading control.

Beyond occupant-focused models, Sun et al. [17] explored a real-time façade monitoring approach based on deep learning and image recognition to assess window opening behavior. Targeted at severe cold regions, the system used video-based monitoring from public street element cameras and achieved an accuracy of 97.79% in detecting window states. The method, built with OpenCV and CNNs, allowed for large-scale, non-intrusive monitoring, offering valuable insights into user behavior for improving simulation accuracy and informing energy-saving strategies.

In addition to comfort and monitoring, the integration of energy generation into responsive façade systems was addressed by Boloria et al. [18]. Their study examined a real-time adaptive building-integrated photovoltaic (BIPV) system that optimized both visual comfort and energy generation through multi-objective evolutionary computing. Compared with static BIPV configurations, the adaptive system demonstrated higher performance in managing daylight, mitigating glare, and enhancing renewable energy yield within the climatic context of Sydney, Australia. This research highlights the potential of AI-driven adaptive façades to address multiple performance objectives simultaneously.

Huchuk et al. [19] developed a Model-Based Predictive Control (MPC) system for interior roller shades using EnergyPlus simulations to compare its effectiveness with a conventional reactive shading system. The MPC algorithm was designed to optimize shading operations based on weather predictions and real-time indoor conditions. Their results indicated that MPC could significantly reduce energy consumption, achieving 12% heating savings, 49% cooling savings, and 54% lighting energy savings compared to conventional shading control strategies.

Manzan et al. [20] proposed an optimization approach to shading control, integrating genetic algorithms with EnergyPlus simulations. The study focused on optimizing the geometry and placement of fixed external shading devices in combination with internally controlled venetian blinds. The GA algorithm was employed to minimize HVAC and lighting energy consumption while maintaining visual comfort metrics such as daylight availability and glare probability.

Yun et al. [21] evaluated visual comfort and energy consumption using ten different model-based control (MBC) strategies for Venetian blinds. These systems use algorithms and predictive models to control shading devices based on inputs such as the time of day, weather forecasts, and occupancy patterns. The goal of these systems is to optimize energy efficiency and occupant comfort by anticipating and responding to changing conditions. Their results indicated that vertical eye illuminance (E_v) was a superior measure of glare compared to DGP, aligning with earlier research [22]. Additionally, Konstantzos et al. [23] found that correlations involving vertical eye illuminance outperformed both DGP and work plane illuminance in assessing glare.

The findings of Yun et al. [21] and Konstantzos et al. [23] indicate that E_v is a superior metric for glare assessment, yet many automated systems still rely on outdated discomfort glare indices (e.g., DGP and DGPs) that may not accurately reflect user experience under varying daylight conditions. E_v measures the amount of light reaching the eye from a vertical plane, influenced by factors such as building design, window dimensions, shading features, and external conditions. Standard values for E_v in buildings are typically set at 500 lux to ensure comfort, with increases to 1000 lux for optimal visual performance [24]. However, these values can be adjusted based on specific contexts and building uses. In cases where sunlight is present [25], multiple metrics, including Task Illuminance (E_t), are recommended. E_t refers to the overall amount of light falling on a surface from all directions and is used to evaluate general lighting conditions in a space, ensuring sufficient light for various tasks and activities. Standard values for E_t can vary depending on the application, but general guidelines suggest maintaining adequate levels to support the intended use of the space, such as 300–500 lux for general office work [26,27].

3. Methodology

The paper's methodology is structured into three primary phases depicted in Fig. 2: Data Generation, Model Development, and Optimization. The main goal is to develop an intelligent control system for office environments that optimizes daylight usage while enhancing visual comfort.

3.1. Data generation

A simulated shared office space was designed in Grasshopper with dimensions of 10 m in length, 5 m in width, and 3 m in height, following the AS/NZS:4442 standards [28]. The office's south-facing wall was fully glazed, featuring a 90 percent window-to-wall ratio and a 0.5-m structural frame depth to support the façade system. To minimize the impact of external shading elements like trees or nearby buildings, the office was assumed to be located on the top floor of a high-rise building. As a shading system, a non-conventional adaptive façade was designed and installed across the entire window. The façade featured a polygonal framework subdivided into four equilateral triangles. These triangles could fold up to 90 degrees along a central symmetrical line and ascend at angles of 45 or 60 degrees, ultimately converging at the polygon's center. To enhance its responsiveness, an attractor point was incorporated in this way that panels closer to the attractor point opened wider, while those farther away remained more closed. This design enabled the façade to respond dynamically to user location preferences and external daylight conditions, ensuring an occupant-centered approach. Fig. 3(a) presents a visual representation of the office model with its façade and Fig. 3(b) shows a close-up of the façade panels. This dynamic folding mechanism enhanced both functionality and aesthetics while allowing the building to adapt to changing daylight conditions, regulating the amount of light entering the office.

Daylight simulations are carried out using the Honeybee plugin for Grasshopper, which utilizes the Radiance engine to provide a detailed analysis of daylight dynamics. Standard parameters are used for the office materials, Table 1 shows the details of the material properties of each building element.

To generate the dataset, the EPW file of nine American cities was used as a representation of different climates to increase the generalizability of the model. The cities include: Seattle, Los Angeles, Phoenix, Denver, Minneapolis, Chicago, New York, Houston, and Atlanta. Two daylight metrics were analyzed: Task Illuminance (E_t) and Vertical Eye Illuminance (E_v). E_t and E_v were assessed via "HB Point-In-Time Grid-Based" and "LB View Percent" view ratios. Fig. 4 shows three samples of the simulations.

For the automation and randomization of data generation, the locations, attractor point positions, and time (hour, day, and month) were randomly varied within a specified range. The time was randomized between 9 AM and 5 PM for all days of the year. Environmental data, including dry-bulb temperature, relative humidity, wind speed, and cloud cover, were collected from EPW files. Using the Zhang and Huang solar model [29], the solar irradiance for each specific hour was calculated. In real-world applications, these environmental data would be collected by external sensors instead of EPW files.

To record the data and develop a surrogate model, data from 500 sensors in the simulation were collected. Each sensor represents a single data point. The features include the distance of a sensor to the south, east, north, and west walls; the distance to the attractor point; the angle between the sensor and the attractor point; and the sky condition at that particular hour. The targets are E_v and E_t of each sensor. Table 2 summarizes all the features and targets of the dataset.

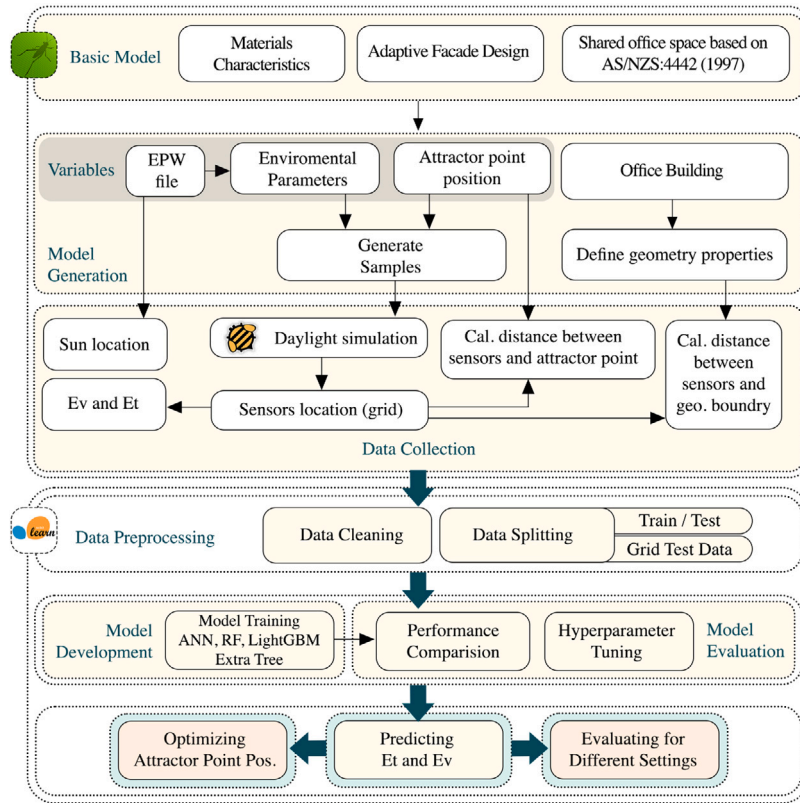


Fig. 2. Methodology overview.

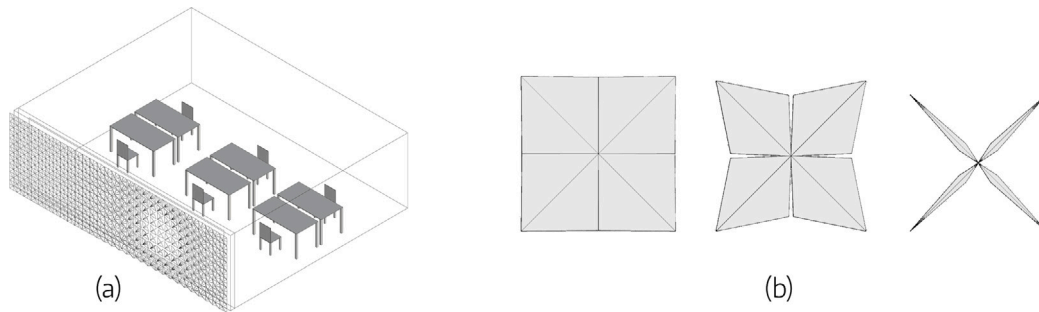


Fig. 3. (a) 3D overview of the office (b) Close-up of the façade panel.

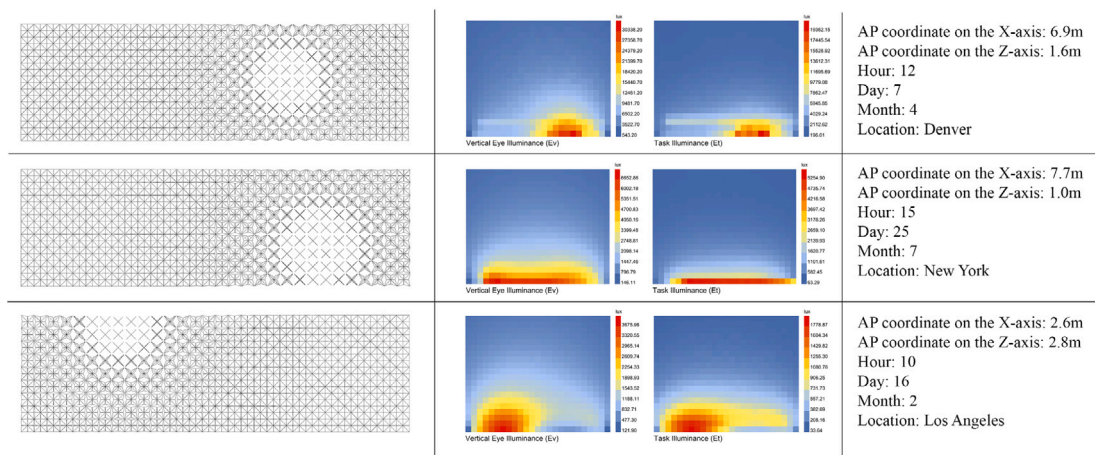


Fig. 4. Three E_v and E_t samples from Ladybug simulation.

Table 1
Radiance materials characteristics.

Parameters	Assigned value(s)
Building elements	Wall: Diffuse Reflectance: 0.5, Specularity: 0
	Floor: Diffuse Reflectance: 0.2, Specularity: 0
	Ceiling: Diffuse Reflectance: 0.8, Specularity: 0
	Structural frame of the window: Diffuse Reflectance: 0.3, Specularity: 0.1
Window	r-diff-ref: 0.3
	g-diff-ref: 0.3
	b-diff-ref: 0.3
	diff-trans: 0.5
	spec-trans: 0.1
	rough: 0.1
Shading	Red diffusion reflectance: 0.25
	Green diffusion reflectance: 0.56
	Blue diffusion reflectance: 0.35
	Specular reflection: 0
	Diffusion transmission: 0.39
	Specular transmission: 0.05
	Roughness: 0.2

Table 2
Dependent and independent variables.

Outputs	Daylight metrics	Vertical eye illuminance (E_v) Task illuminance (E_t)
Inputs	Geometric data	Distance of the sensor to the south wall Distance of the sensor to the east wall Distance of the sensor to the west wall Distance of the sensor to the north wall Distance of the sensor to the attractor point Angle between the sensor and the attractor point Attractor point coordinate on the X-axis Attractor point coordinate on the Z-axis
	Environmental data	Dry-bulb-temperature Relative-humidity Wind-speed Latitude Longitude Elevation Cloud cover Direct-wea Diffuse wea Global wea Azimuth Altitude

3.2. Model development

Three machine learning algorithms – Artificial Neural Networks (ANN), Random Forest (RF), Light Gradient-Boosting Machine (LightGBM) and Extra Tree – were chosen due to their strong ability to capture and model complex nonlinear relationships in daylight predictions [30]. Each algorithm was applied and evaluated in developing surrogate models for predicting Task illuminance, and Vertical Eye illuminance. Given that different algorithms exhibit varying characteristics, this comparative approach ensures that the most suitable model is selected based on performance in different scenarios.

Artificial Neural Networks (ANN) is a core component of supervised learning in deep learning, designed to recognize patterns and relationships within complex datasets. ANN adaptability in handling non-linear relationships between input variables (e.g., time of day, weather conditions, building geometry) and output variables (illuminance levels) makes it a reliable alternative to traditional daylight modeling approaches like scale models, analytical formulas, or simulation software [31]. This study utilizes a feed-forward backpropagation neural network, a type of artificial neural network where data moves in one direction – from input to output – without looping back. The input layer, in the initial step before feature engineering, contains 13 neurons, each representing a feature, while the output layer has 2 neurons corresponding to the outputs, E_v and E_t . The number of hidden

layers varies during model training and grid search to identify the optimal parameters.

Random Forest (RF) is a popular ensemble learning technique commonly applied to supervised learning tasks, especially regression problems, due to its ease of use, reliability, and high predictive accuracy [32]. It constructs multiple decision trees during training and combines their outputs to enhance accuracy and reduce overfitting. RF's ability to handle high-dimensional data and capture complex, non-linear relationships between input and output variables makes it a robust alternative to traditional predictive modeling techniques [33]. In this study, RF is applied to model the relationship features and two target outputs, E_v and E_t . The number of trees in the forest and other hyperparameters are optimized through grid search to achieve the best predictive performance.

Light Gradient Boosting Machine (LightGBM) is a gradient boosting framework known for its efficiency in handling large datasets and complex relationships in supervised learning tasks. Unlike traditional boosting methods, LightGBM uses histogram-based learning and leaf-wise tree growth, leading to faster training and better generalization, especially for high-dimensional data [34]. Unlike RF, LightGBM is based on a gradient boosting framework, which sequentially integrates weak learners to build a stronger model, aiming to enhance overall accuracy. It utilizes a highly optimized histogram-based decision tree learning algorithm to minimize memory usage and computational cost. In this study, its hyperparameters, including the number of boosting

iterations, learning rate, and maximum depth, are fine-tuned using grid search to enhance predictive accuracy.

Finally, Extremely Randomized Trees (Extra Trees) is an ensemble learning method that extends the concept of Random Forest by introducing additional randomness in the splitting criteria. Unlike RF, where the best split is selected based on information gain or Gini impurity, Extra Trees randomly selects split points, leading to greater model diversity and improved generalization [35]. This characteristic allows E_t to effectively capture complex relationships between input features and target outputs while reducing variance.

To assess the predictive performance of the models, three commonly used evaluation metrics were employed: Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared (R^2). These metrics provide insight into the accuracy and reliability of the model's predictions by quantifying the difference between predicted and actual values. They can be calculated using Eqs. (1) to (3), respectively.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (1)$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (2)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

Where n is the number of data points, y_i represents the actual value, $\hat{y}_i$ is the predicted value, $\bar{y}$ is the mean of the actual values, $\sum (y_i - \hat{y}_i)^2$ is the residual sum of squares (RSS), measuring the total squared errors in predictions, and $\sum (y_i - \bar{y})^2$ is the total sum of squares (TSS), representing the total variance in the actual values.

MSE calculates the average of the squared differences between predicted and actual values, placing greater weight on larger errors due to squaring. MAE, on the other hand, measures the average absolute differences, making it less sensitive to large deviations. Finally, R^2 measures the proportion of variance in the data that is explained by the model, with values closer to 1 indicating a better fit. A combination of these metrics provides a comprehensive assessment of model performance, allowing for comparison across different approaches and ensuring the selection of the most suitable model for the given task.

To identify the key factors influencing E_v and E_t , we utilized the Extra Trees Regressor to analyze feature importance and determine the most significant predictors for each illuminance type. This method assigns an importance score to each feature based on its impact on reducing variance in the model's predictions, with higher scores indicating greater influence on the target variable. The Extra Trees Regressor was implemented with 50 trees, a maximum depth of 15, and all features considered at each split to enhance interpretability while maintaining computational efficiency.

In addition to predictive modeling, this study employs the Non-dominated Sorting Genetic Algorithm II (NSGA-II) to optimize façade configurations, ensuring sufficient daylight while minimizing glare. NSGA-II, a widely used evolutionary algorithm, efficiently solves complex optimization problems by employing a fast non-dominated sorting approach and an elitist selection strategy to explore a diverse set of potential solutions [36]. In this study, NSGA-II is utilized to determine the optimal positioning of the attractor point on an adaptive façade, aiming to maintain E_t within a suitable range that reduces glare discomfort. Through iterative crossover and mutation operations, the algorithm refines façade configurations, ultimately enhancing daylight performance while improving occupant visual comfort.

3.3. Training of the models and optimization

During the training phase, single-output machine learning models were developed to predict E_v and E_t , as described in previous sections. These models were trained using four selected machine learning algorithms. Single-output models offer a simpler training process, as they

focus on predicting a single target variable at a time, reducing computational complexity. Additionally, they enhance interpretability by allowing a clearer analysis of feature importance and model behavior for each predicted output.

The dataset, after data collection, initially contained 792,000 rows, representing 1584 sets of 500 sensors each. However, due to simulation errors, issues with EPW files, or system crashes, 18 sets had outputs of zero and were removed. As a result, the refined dataset contained 783,000 rows (or 1566 sets). Before training the models, 10% of the sets were set aside for final performance testing. The remaining 90% was split into an 80/20 ratio for training and validation. The reason for reserving a fixed 10% test set was that, in the conventional train-test split, sets are selected randomly and shuffled. This randomization allows for evaluating performance metrics like Mean Absolute Error (MAE) per individual sensor but does not effectively assess performance across an entire sensor grid.

In the first step, four models were trained using 5-fold cross-validation and a randomized grid search to optimize hyperparameters. The performance of each model was then evaluated based on R-squared (R^2), MAE, and Mean Squared Error (MSE), using both the shuffled test set and the structured grid test set. The model that demonstrated the best performance across both test sets was selected for the optimization stage, where it would be used to predict target values.

A 5-fold cross-validation strategy is employed to ensure model stability and generalization. The dataset is randomly shuffled before being split into five subsets, where four are used for training and one for validation in each iteration. This process is repeated five times, and the average performance across folds is used to select the optimal model.

For the E_t and RF models, the number of estimators is optimized within a range of 50 to 200. The tree depth is optimized within a range of None, 10, 20, and 30, where deeper trees capture more complex patterns but may lead to overfitting. The minimum number of samples required to split an internal node is optimized within a range of 2, 5, and 10, while the minimum number of samples required to form a leaf node is optimized within a range of 1, 2, and 4.

For the LightGBM model, the number of estimators is tuned within a range of 50 to 200. The learning rate is optimized within a range of 0.01 to 0.2, where a lower rate ensures stable convergence, and a higher rate accelerates training but may cause divergence. The number of leaves per tree is optimized within a range of 20, 31, and 50, influencing model complexity and overfitting risk. The maximum tree depth is optimized within a range of -1, 10, and 20, where a value of -1 allows unrestricted growth.

For the ANN model, the architecture is optimized by varying the hidden layer sizes among (128, 64, 32), (256, 128, 64), and (64, 32). The regularization term (alpha) is optimized within a range of 0.0001 to 0.01, where a higher value penalizes complexity to prevent overfitting. The initial learning rate is tuned within a range of 0.001 to 0.01, controlling the weight update step size. The maximum number of training iterations is optimized within a range of 200 to 1000, ensuring sufficient learning without excessive computation.

Randomized grid search is applied to efficiently explore the hyperparameter space, reducing computational cost while ensuring near-optimal model selection. The best-performing configurations are then used for feature importance analysis, final evaluation and prediction (see Table 3).

Finally, to determine the optimal attractor point on the adaptive façade, the NSGA-II algorithm was employed via the DEAP library. Every 30 min, an outdoor sensor collected data on environmental factors, which in our dataset included dry-bulb temperature, humidity, cloud cover, and wind speed. Using these four factors, diff-weather, diff-glob, and dir-wea were calculated following the Zhang-Huang method. Additionally, the sun's azimuth and altitude, corresponding to the hour, day, and month, were extracted from the weather file to ensure precise daylight modeling.

Table 3
Hyperparameter tuning for the E_v and E_t prediction models.

Machine learning model	Hyperparameter	Optimization range
Extra Trees (E_t)	n_estimators	50, 100, 200
	max_depth	None, 10, 20, 30
	min_samples_split	2, 5, 10
	min_samples_leaf	1, 2, 4
Random Forest (RF)	n_estimators	50, 100, 200
	max_depth	None, 10, 20, 30
	min_samples_split	2, 5, 10
	min_samples_leaf	1, 2, 4
LightGBM (LGB)	n_estimators	50, 100, 200
	learning_rate	0.01, 0.1, 0.2
	num_leaves	20, 31, 50
	max_depth	-1, 10, 20
Artificial Neural Network (ANN)	hidden_layer_sizes	(128,64,32), (256,128,64), (64,32)
	alpha	0.0001, 0.001, 0.01
	learning_rate_init	0.001, 0.01
	max_iter	200, 500, 1000

To determine the optimal placement of the Attractor Point (AP), the optimization sought to maximize predicted E_t (illuminance) while simultaneously minimizing E_v (glare and excessive light exposure). The optimization was constrained by ensuring that E_v remained below 1000, preserving occupant comfort.

Each candidate solution in the NSGA-II population represented a potential AP coordinate (length and width), with initial values randomly generated using NumPy's uniform distribution function. The algorithm systematically explored feasible AP placements that would enhance illumination while maintaining visual comfort.

To maintain realistic and practical solutions, mutation and crossover operations in DEAP were adapted to enforce boundary constraints, ensuring that AP coordinates remained within spatial limits. The mutation process introduced Gaussian perturbations, while crossover operations blended parent solutions to explore new AP configurations. Each individual solution was evaluated using two separately trained Extra Trees Regressor (ETR) models—one for predicting E_t and another for E_v . These models incorporated dynamically computed spatial parameters such as distances and angles relative to fixed sensor locations, as well as the environmental features extracted from climate datasets.

The genetic algorithm was executed with a population size of 50 over 50 generations, allowing extensive exploration of the search space. Unlike a single-objective optimization, where a single AP placement would be selected, this approach generated a Pareto front—a set of non-dominated solutions representing optimal trade-offs between E_t and E_v .

Finally, Fig. 5 illustrates the complete workflow of the proposed control algorithm. The left section depicts the machine learning model development process, which includes data preprocessing, hyperparameter tuning through grid search and cross-validation, and final model selection based on performance metrics. The right section outlines the real-time control loop for the adaptive façade system. During occupied hours, the system collects environmental and interior data, encodes the attractor point as an input, and uses the trained machine learning models to predict E_t and E_v . These predictions are passed to a genetic algorithm-based optimizer, which identifies the optimal attractor point coordinates that maximize task illuminance while maintaining vertical eye illuminance below 1000 lux. The system then updates the façade accordingly. If the optimized E_t falls below the acceptable threshold, artificial lighting is activated. This feedback loop advances in hourly time steps, ensuring continuous and responsive daylight management throughout the day.

4. Results

The entire training process was conducted on a system with 16 GB DDR3 RAM, a 2.5 GHz Intel Core i7 CPU, and an NVIDIA GeForce GTX 960M GPU. The data generation process took 48 h. The Table 4 shows the key statistics metrics of the dataset.

4.1. Model evaluation

The results for the Task Illuminance are presented in Table 5 highlighting the best model parameters and their corresponding performance metrics. The ANN model achieved the highest R^2 score (0.9888) and the lowest MSE (30,456.72), demonstrating its strong predictive capabilities. However, its MAE (60.59) was higher than that of the Extra Trees model (53.93). Since MAE is a crucial metric for assessing model robustness – being less sensitive to outliers and providing a more interpretable measure of average error – it was chosen as the primary selection criterion. Based on this, the Extra Trees model was deemed the most suitable, as it consistently produced the smallest average prediction errors while maintaining a high R^2 score (0.9825).

The Random Forest model performed comparably to Extra Trees, with an R^2 of 0.9823 and an MAE of 60.05, slightly higher than that of Extra Trees. However, LightGBM exhibited a significantly higher MAE (95.84) despite having an R^2 of 0.9820, suggesting that it struggled with consistent accuracy. This further supports the selection of Extra Trees as the best model for predicting daylight illuminance.

To further validate the models, a grid Testing Set was used to assess how well they capture the spatial structure of the sensor grid, with results shown in Table 6. The mean R^2 value in this table represents the average predictive performance across 156 test sets that were withheld before training. The Extra Trees model again demonstrated superior performance, achieving the highest mean R^2 (0.95) and the lowest standard deviation (0.06). This indicates that Extra Trees not only outperformed other models on average but also exhibited more consistent results, reinforcing its robustness for real-world applications. The Random Forest model also performed well, with a mean R^2 of 0.94, but it exhibited a higher standard deviation (0.10), indicating slightly less consistency in predictions compared to Extra Trees. Meanwhile, the ANN model, despite achieving high accuracy in the shuffled testing set, showed a drop in performance on the grid-based evaluation, with an R^2 of 0.87 and a standard deviation of 0.89, suggesting higher variability. The LightGBM model struggled significantly, achieving only 0.28 mean R^2 with a standard deviation of 6.19, indicating an unstable and unreliable performance in capturing spatial patterns. The best and worst predicted grids for E_t can be found in Figs. A.1 and A.2 in the Appendix.

Beyond raw performance metrics, the Extra Trees model offers an additional advantage in terms of interpretability discussed in following section. Unlike ANN, which functions as a black-box model, tree-based methods such as Extra Trees and Random Forest allow for feature importance analysis.

The results of the Vertical Task Illuminance are presented in Table 7, highlighting the best parameters of the model and their corresponding performance metrics. Similar to the Task Illuminance, the Extra Trees

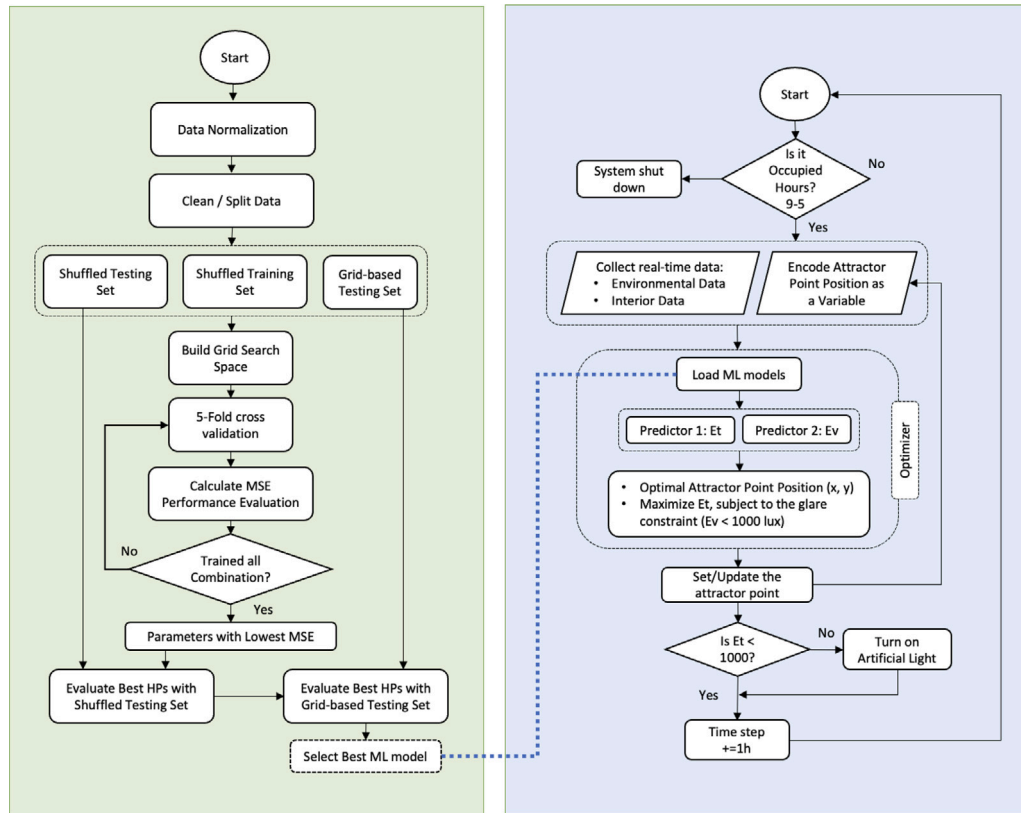


Fig. 5. Algorithm pipeline from model development to real-time optimization of the attractor point.

Table 4

Key statistical metrics of the dataset.

Features and Targets	mean	std	min	max	median
SP-Soth-Dis	4	2.31	0.2	7.8	4
SP-East-Dis	5	2.88	0.2	9.8	5
SP-North-Dis	4	2.31	0.2	7.8	4
SP-West-Dis	5	2.88	0.2	9.8	5
SP-Ap-Dis	5.98	2.31	0.4	12.84	6.07
Angle	12.04	11.04	0	79.70	9.23
AP-Length	5.15	2.88	0	10	5.3
AP-Width	1.51	0.98	0	3	1.5
Dir-wea	488.71	339.95	0	1010	577.5
Diff-wea	141.97	86.97	0	417	131.5
glob-wea	483.01	316.94	0	1053.60	494.62
Total-Sky-cover	4.93	3.87	0	10	5
Altitude	39.17	18.35	4.732692	73.72	39.93
Azimuth	185.01	49.65	105.230554	276.10	180.00
Dry bulb temperature	17.00	9.07	-10.5	40	16.7
Relative humidity	55.56	22.97	5	100	53
Wind speed	4.41	3.50	0	12.2	3.6
Latitude	39.16	5.84	29.72	47.61	39.72
Longitude	-103.59	15.25	-122.338	-73.96	-104.9
Elevation	336.23	504.59	2	1645	181
E_t	867.42	1639.38	0	25 410.9	286.5
E_v	2216.92	3182.40	0	52 647.0	1226.3

Table 5

Best model parameters and performance for E_t (shuffled testing set).

Model	Best parameters	R ² Score	MAE	MSE
Extra Trees	'n_estimators': 100, 'min_samples_split': 5	0.9825	53.93	47 487.43
Random Forest	'n_estimators': 200, 'min_samples_split': 2	0.9823	60.05	47 984.62
LightGBM	'n_estimators': 200, 'learning_rate': 0.1	0.9820	95.84	48 682.10
ANN	'hidden_layer_sizes': (128, 64, 32), 'alpha': 0.1	0.9888	60.59	30 456.72

Table 6Performance comparison for E_t (grid testing data).

Models	Mean R^2	Std
Extra Tree	0.95	0.06
Random Forest	0.94	0.10
LightGBM	0.28	6.19
ANN	0.87	0.89

model demonstrated strong predictive capabilities, achieving the highest R^2 score (0.99) and the lowest MAE (90.29). Although the ANN model also achieved a high R^2 score (0.99), its MAE (139.21) was significantly higher, making it less favorable for practical applications. Given that MAE is a key metric for assessing model robustness, Extra Trees was deemed the most suitable model due to its ability to consistently produce the smallest average prediction errors.

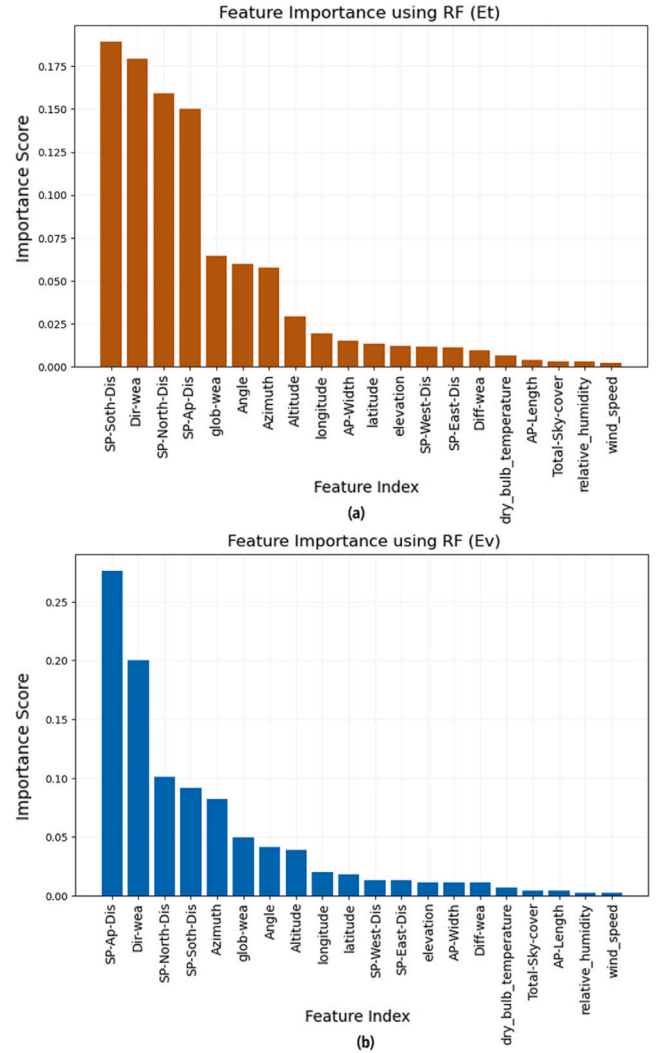
The Random Forest model performed comparably to Extra Trees, with an R^2 score of 0.98 and an MAE of 108.74. However, LightGBM, despite achieving an R^2 of 0.98, exhibited the highest MAE (170.93), indicating a poorer predictive accuracy. This further supports the selection of Extra Trees as the best model for predicting vertical illuminance.

To further assess the model's generalizability, a Grid Testing Set was used to evaluate how well the models capture the spatial structure of the sensor grid, with results shown in Table 8. The Extra Trees model again demonstrated superior performance, achieving the highest mean R^2 (0.95) with a low standard deviation (0.06), indicating consistent predictive accuracy across different spatial configurations. The Random Forest model performed similarly, with a mean R^2 of 0.94 and a slightly higher standard deviation (0.08), suggesting slightly more variability in predictions. The ANN model maintained strong predictive performance with a mean R^2 of 0.89 and a standard deviation of 0.11, reflecting relatively stable performance across different test sets. However, LightGBM exhibited the weakest performance, with a mean R^2 of 0.66 and a significantly high standard deviation of 1.81, indicating high variability and unreliable spatial predictions. These findings reinforce the conclusion that the Extra Trees model is the most suitable for predicting vertical illuminance due to its balance of accuracy and stability. The best and worst predicted grids for E_v can be found in Figs. A.3 and A.4 in the Appendix.

4.2. Feature importance

Feature importance analysis provides insights into the critical parameters influencing predictive accuracy for Vertical Eye Illuminance (E_v) and Task Illuminance (Et) using the Extra Trees Regressor, the best-performing model identified in this paper.

For vertical eye illuminance (E_v) in Fig. 6(a), spatial parameters emerged as the most influential predictors. The distance between the sensor and the attractor point (AP distance) exhibited the highest impact, emphasizing that proximity to specific façade features significantly governs vertical illuminance levels. Direct solar radiation (Dir-wea) was identified as the second most significant predictor, underscoring the critical role of direct sunlight in determining vertical illuminance. Furthermore, sensor distances from the north and south walls (SP-North-Dis and SP-South-Dis) were highly influential, reinforcing the importance of sensor orientation concerning daylight ingress. Conversely, features related to general meteorological conditions such as cloud cover, humidity, or wind speed exhibited relatively minor influence, underscoring that vertical eye illuminance is predominantly governed by direct sunlight and spatial positioning rather than ambient environmental factors. In contrast, task illuminance (E_t), as shown in Fig. 6(b), was primarily influenced by sensor placement relative to the south wall (SP-South-Dis), indicating that horizontal daylight distribution significantly depends on the sensor's position relative to the southern façade. Direct solar radiation (Dir-wea) remained critically

**Fig. 6.** Feature importance for (a) E_t (b) E_v .

important, confirming its substantial role in horizontal daylight levels. Other influential variables included the sensor's distance from the north wall, attractor point proximity, and angle of incidence, suggesting that task illuminance is notably affected by both spatial geometry and the interaction between direct and diffuse daylight. Unlike vertical illuminance, which predominantly responds to direct sunlight and spatial configuration, task illuminance displays sensitivity to a broader range of daylight factors, including the interplay between façade geometry and daylight characteristics.

These findings align with previous research, such as the study by Luo et al. [16], highlighting the dominance of spatial factors and direct solar radiation in affecting interior visual comfort metrics, particularly in automated shading and daylight control scenarios.

4.3. Optimization

Four random samples were chosen to demonstrate the optimization's effectiveness under different conditions. As shown in Table 9, the selected cases represent different cities, dates, weather conditions, and times. Additionally, two user locations were chosen as samples for optimization. Table 10 shows the coordinates of the users in the room.

Each plot presents the Pareto front for two different sensors (Sensor 1 and Sensor 3), marked in blue and red, respectively. Across all four

Table 7
Best model parameters and performance for E_v (shuffled testing set).

Model	Best parameters	R ² Score	MAE	MSE
Extra Trees	'n_estimators': 200, 'min_samples_split': 2, 'min_samples_leaf': 1, 'max_depth': 30	0.99	90.29	97 652.52
Random Forest	'n_estimators': 200, 'min_samples_split': 2, 'min_samples_leaf': 1, 'max_depth': 30	0.98	108.74	107 269.37
LightGBM	'num_leaves': 31, 'n_estimators': 200, 'max_depth': 10, 'learning_rate': 0.2	0.98	170.93	105 511.28
ANN	'max_iter': 500, 'learning_rate_init': 0.001, 'hidden_layer_sizes': (128, 64, 32), 'alpha': 0.001	0.99	139.21	86 298.24

Table 8
Performance comparison for E_v (grid testing data).

Models	Mean R ²	Std
Extra Trees	0.95	0.06
Random Forest	0.94	0.08
LightGBM	0.66	1.81
ANN	0.89	0.11

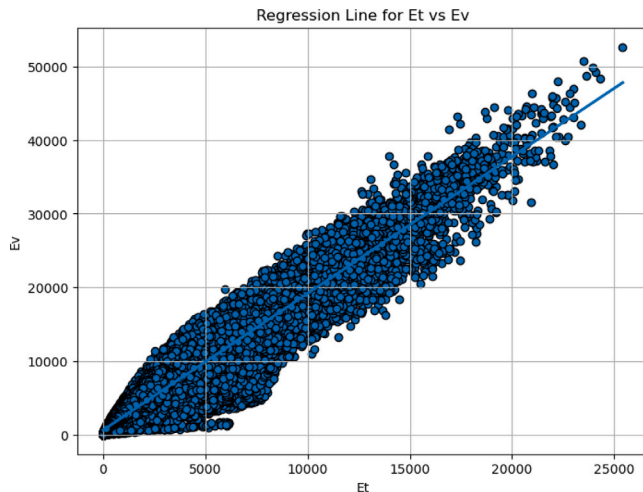


Fig. 7. Scatter plot and regression line for E_t vs. E_v .

locations, a clear Pareto front is observed, illustrating the correlation between E_t and E_v (Fig. 7). The Pareto plots demonstrate that the system effectively optimizes the building façade by maximizing E_t while minimizing E_v . The optimal value for each sample is constrained to a point where E_v remains below 1000 lux, while E_t is maximized (Fig. 8).

5. Discussion

This paper developed a real-time façade control algorithm using a machine learning-based surrogate model trained with simulations from Radiance and Ladybug Tools. The model was designed to predict daylight metrics using typical environmental sensor input in real-world applications while providing high-accuracy results. By leveraging this

model as a control algorithm, an optimization function can be integrated to dynamically adjust the façade based on user location, ensuring indoor lighting remains within a standard range. This approach minimizes glare, enhances visual comfort, and improves overall functionality in office environments. The results showed that the Extra Trees Regressor achieved the highest accuracy, demonstrating the model's ability to provide real-time predictions of E_v and E_t for specific locations within the room. The comparison between actual and predicted illuminance values in the worst-predicted sets highlights notable differences in how models capture spatial lighting distribution. While Extra Trees and Random Forest models show reasonable performance in predicting task illuminance, their predictions for vertical eye illuminance exhibit significant deviations from the actual values, particularly in intensity distribution. The Extra Trees model, for instance, overestimates illuminance levels, producing a broader and more intense spread compared to the actual distribution. Similarly, the Random Forest model captures some spatial patterns but introduces blocky artifacts, reflecting the inherent partitioning nature of decision trees.

The ANN model struggles to maintain spatial coherence in both E_t and E_v , generating highly dispersed and unrealistic illuminance patterns. In the case of E_v , it fails to accurately capture peak values, leading to significant underestimations and even negative values in certain regions. This suggests that the model lacks sufficient spatial awareness, possibly due to an inadequate representation of spatial dependencies in the training process. Likewise, the LightGBM model shows poor generalization, especially for vertical illuminance, with irregular patterns, artifacts, and negative predictions. These inconsistencies highlight the need for additional constraints to prevent physically unrealistic outputs in this model.

The insights gained from feature importance analysis have significant implications for the design of adaptive shading control systems and algorithms. Sensor placement and façade geometry emerge as key factors in optimizing daylight management, ensuring occupant visual comfort. These findings highlight the necessity of incorporating dynamic shading strategies that respond to spatial positioning and direct solar exposure to regulate indoor lighting conditions effectively.

For E_v , results suggest that façade geometry, sensor distance from reference points, and direct solar exposure should be prioritized. Strategies such as dynamic façades, shading devices, and precise sensor positioning can significantly improve vertical daylight distribution, minimizing glare while maximizing natural light penetration.

In the case of E_t , the most critical predictors include global solar radiation, direct sunlight, and spatial positioning. This suggests that workspace design should account for factors such as desk placement, shading controls, and daylight-responsive artificial lighting systems to maintain optimal task illuminance. While both E_v and E_t are strongly

Table 9

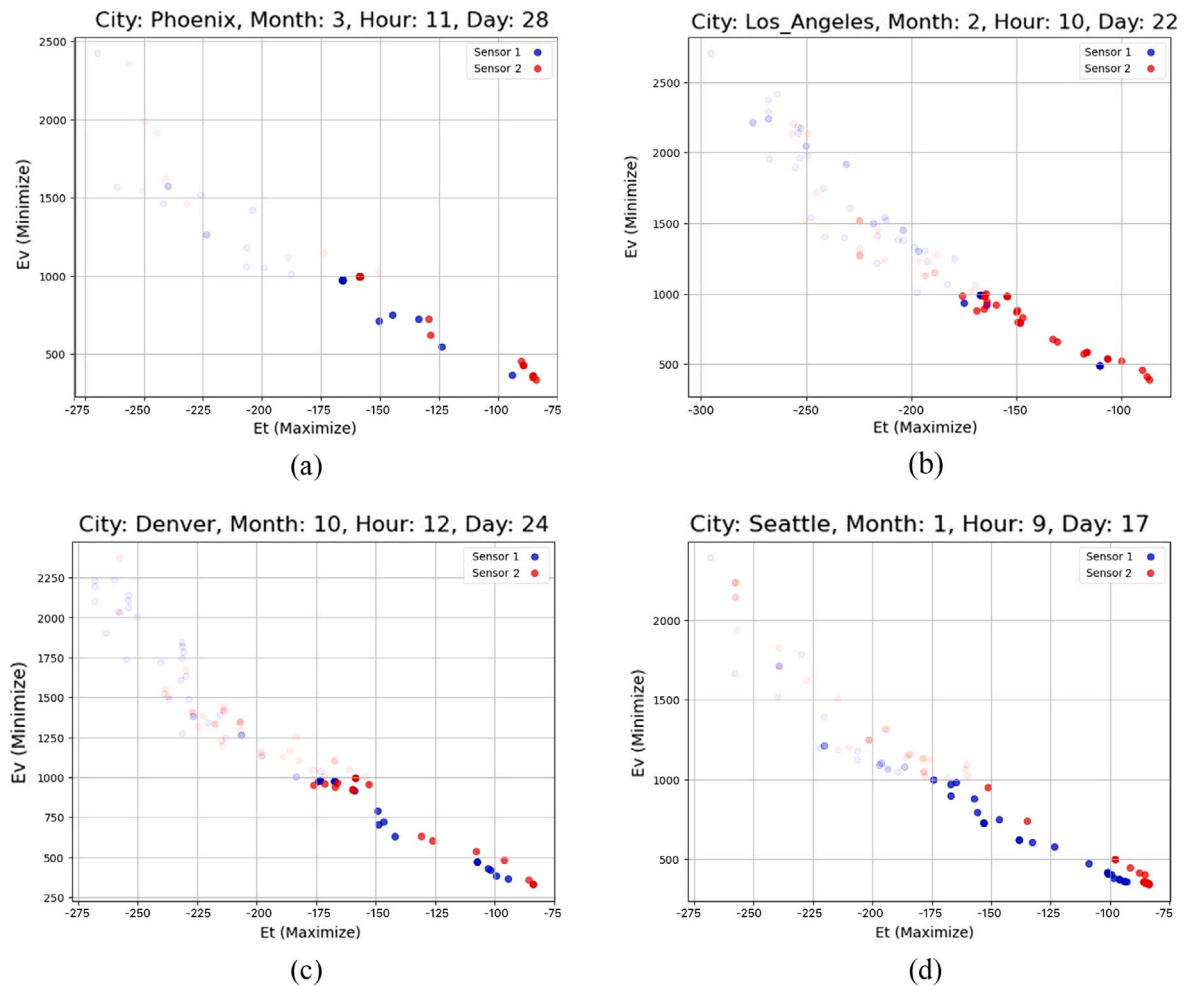
Four selected random conditions for optimization.

City	Month	Day	Hour	Total-Sky-Cover	Dry bulb temperature	Relative humidity	Wind speed
Phoenix	3	28	11	2	21.7	31	2.6
Los Angeles	2	9	10	0	12.8	36	1.5
Denver	10	24	12	10	3.9	79	7.7
Seattle	1	17	9	8	8.4	82	3.6

Table 10

Two selected random user locations in the room for optimization.

User	Distance to south wall (m)	Distance to East wall (m)	Distance to North wall (m)	Distance to West wall (m)
1	3.8	3	4.2	7
2	4.2	7	4.2	3

**Fig. 8.** Pareto plot optimization of four random samples (a) Phoenix (b) Los Angeles (c) Denver (d) Seattle.

influenced by direct solar radiation and spatial positioning, task illuminance exhibits a greater dependency on global radiation and surface angles, whereas vertical illuminance is more affected by façade geometry and attractor distances. The optimization function demonstrated that the model successfully optimized the façade based on input parameters. Although the optimization function helps prevent glare, the task illuminance is not within the standard range, requiring either a better façade design or the use of artificial lighting. Another key finding of this paper is that testing the model with data from cities not included

in the training dataset demonstrated its ability to generalize, highlighting its potential for application in façade control across different locations. Unlike previous shading control systems that rely solely on rule-based approaches, which sometimes fail to adjust based on real-time sky conditions, this model integrates real-time daylight prediction using environmental inputs and location data for a specific hour. This approach enhances daylight prediction in realistic scenarios, ensuring more adaptive and responsive shading control.

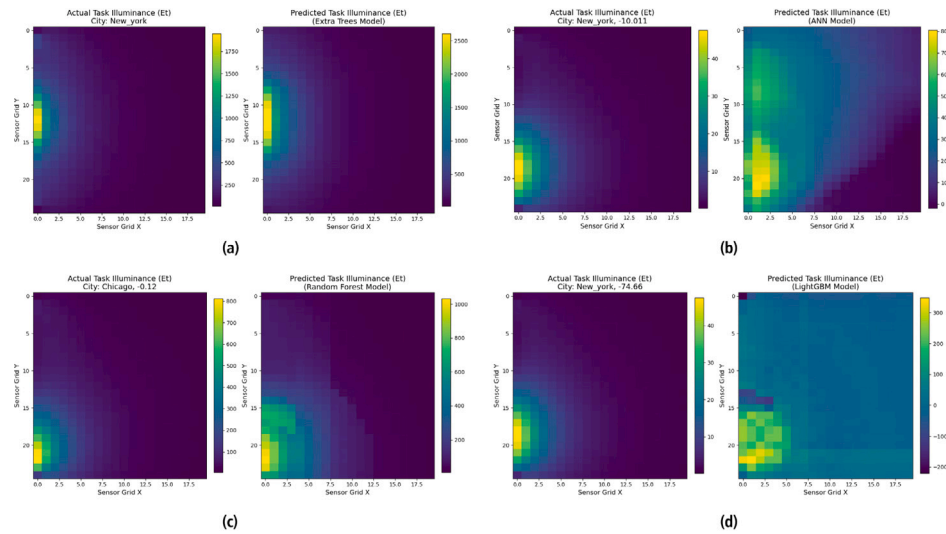


Fig. A.1. Worst predicted grids for E_t of four different machine learning models (a) Extra Tree (b) ANN (c) Random Forest (d) LightGBM.

6. Conclusion

This paper developed an AI-driven real-time shading control system using machine learning-based surrogate models to optimize daylighting and visual comfort in office environments. One of the key advancements in this work is the generalizability of the proposed model. In contrast to previous studies [15,16], which relied on analytical models constrained to simpler shading geometries, our method leverages Genetic Algorithms, enabling it to effectively handle and optimize complex and non-standard façade designs. Additionally, our approach adopts a multi-objective optimization framework, simultaneously optimizing both E_v and E_t to enhance visual comfort and daylighting performance. Another significant contribution is that our model was trained on numerical weather data from multiple cities, improving its generalizability across diverse climate conditions, which can be an improvement compared with work by Biloria et al. and Sun et al. [17,18]. This allows the proposed system to be applicable to similar façade configurations in a wide range of environmental contexts. The results confirm that AI-driven optimization can significantly enhance adaptive façade systems compared to traditional rule-based approaches. In terms of limitations, one limitation of this paper is that the developed model was trained exclusively on simulated data rather than actual real-world measurements. While simulations offer a controlled environment and facilitate the systematic analysis of various conditions, real-world deployment may introduce additional variability stemming from unpredictable occupant behavior, sensor inaccuracies, and environmental fluctuations. Another limitation arises from the geographical constraint of the dataset. The current model's training data were limited to a few selected locations within the United States. To further strengthen the model's robustness, future research should incorporate data from a broader range of locations with diverse climatic conditions. Additionally, due to constraints related to the scope, timeframe, and hardware resources available for dataset generation, the current model did not consider furniture, computer screen positioning, and occupant presence within the simulated spaces. Future studies could significantly enhance the model's practical applicability by integrating these critical factors into the training data. Furthermore, the adaptive façade system developed in this paper relies exclusively on environmental parameters for adjustment, omitting user preference or manual intervention capabilities. Given that user preferences and manual overrides are essential features of smart systems for occupant comfort and satisfaction,

incorporating these elements represents an important area for future research and development. Finally, it should be noted that the current façade design, utilizing attractor points and paneling, was used as a representative sample to demonstrate the proposed method. However, the developed model is not yet generalized to other types of façade configurations. Future work should consider different kinds of façade designs to enhance the generalizability and applicability of the system. An additional suggestion for future research involves integrating artificial lighting systems that activate when natural lighting levels are insufficient. Such a hybrid lighting approach could ensure consistent illuminance levels, thereby enhancing occupant visual comfort and productivity.

CRediT authorship contribution statement

Mohammad Tabatabaei Manesh: Writing – Review & Editing, Validation, Methodology, Data Analysis, Visualization, Data Curation. **Mohammad Rajaian Hoonejani:** Writing – Review & Editing, Validation, Methodology, Data Analysis, Visualization. **Samira Ghafari Gousheh:** Writing – Review & Editing, Visualization, Methodology, Data Curation, Conceptualization. **Alireza Abdolmaleki:** Writing – Review & Editing, Conceptualization. **Arman Nikkhah Dehnavi:** Conceptualization, Methodology, Investigation. **Atefeh Shahrashoob:** Visualization.

Declaration of competing interest

The authors declare that there is no conflict of interest regarding the publication of this manuscript.

Appendix

See Figs. A.1–A.4.

Data availability

The data that support the findings of this paper are openly available in GitHub at https://github.com/Mhtaba76/ML_Facade_Control_System.

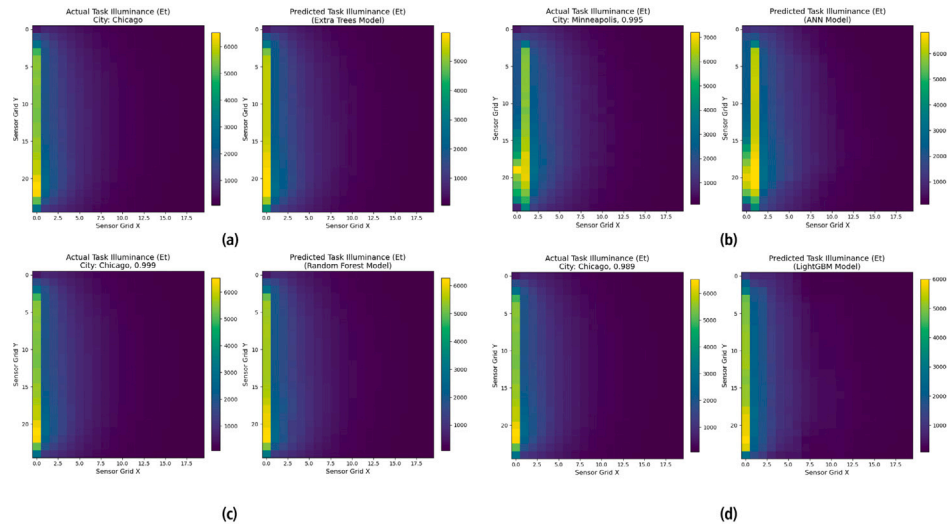


Fig. A.2. Best predicted grids for E_t of four different machine learning models (a) Extra Tree (b) ANN (c) Random Forest (d) LightGBM.

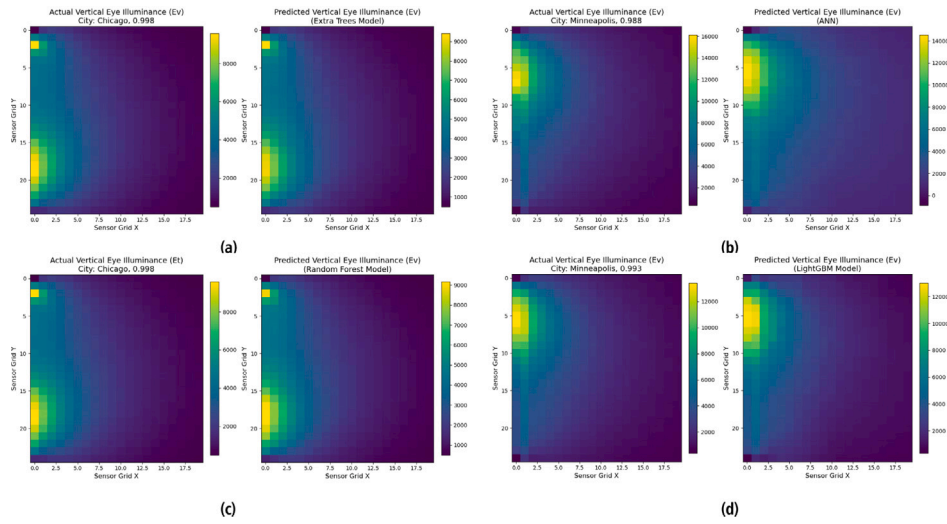


Fig. A.3. Best predicted grids for E_v of four different machine learning models (a) Extra Tree (b) ANN (c) Random Forest (d) LightGBM.

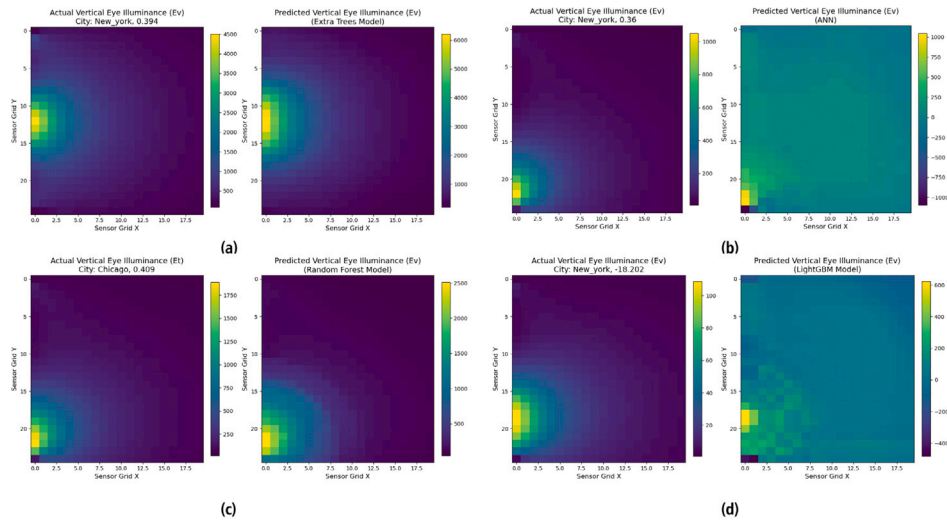


Fig. A.4. Worst predicted grids for E_v of four different machine learning models (a) Extra Tree (b) ANN (c) Random Forest (d) LightGBM.

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